

3. **Sleeping sickness is caused by**

- (a) Giardia (b) Leishmania
(c) Trypanosoma (d) Entamoeba

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

Sleeping sickness (Human African Trypanosomiasis) is caused by Protozoan Parasites transmitted by infected Tsetse flies. It is endemic in Sub-Saharan Africa. Human African Trypanosomiasis takes two forms, depending on the sub-species of the infecting parasite—

- (i) Trypanosoma brucei gambiense
(ii) Trypanosoma brucei rhodesiense

4. **From the following pairs, find the one which is correctly matched.**

- (a) Scurvy - Thiamine (b) Tuberculosis - ATS
(c) Tetanus - BCG (d) Malaria - Chloroquine
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

BCG vaccine is given to prevent tuberculosis. Erabrine, chloroquine etc. are used for the treatment of Malaria. Scurvy is a disease caused by the deficiency of vitamin C, whereas thiamine is known as vitamin B₁.

5. **Which statement is not true regarding malaria?**

- (a) It is a disease caused by insect parasite
(b) It is spread by the mosquitoes
(c) It spreads mostly in swampy region
(d) For its treatment chloroquine is used

38th B.P.S.C. (Pre) 1992

Ans. (a)

Malaria disease is caused by plasmodium parasite which is a protozoa. Thus option (a) is not true. Other three options are correct.

6. **In the environment, plastics will not be decomposed by the action of**

- (a) bacteria (b) saprophyte
(c) parasite (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

Plastic is non-biodegradable and does not decomposed by the action of Bacteria, Saprophytes, Parasite.

7. **Which of the following is a fungal poultry disease?**

- (a) Moniliasis (b) Fowl pox
(c) Ranikhet (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (a)

Moniliasis is a disease of the alimentary tract of chickens, turkey, and sometimes other birds. The cause is the fungal yeast, Candida albicans, and the condition is seen worldwide. Morbidity and mortality is usually low. The route of infection is normally oral.

V. Genetic Disorders

1. **'Thalassemia' is a hereditary disease affecting**

- (a) blood (b) lungs
(c) heart (d) kidney
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (a)

Thalassemia is an inherited blood disorder caused when the body does not make enough of a protein called haemoglobin. When there is not enough haemoglobin, the body's red blood corpuscles (RBCs) do not function properly and they last shorter periods of time, so there are fewer healthy red blood corpuscles traveling in the bloodstream. So this condition can cause anemia.

2. **In respect of women, men can suffer more of colour blindness because-**

- (a) They pass more through mental tensions
(b) They live out of home relatively more
(c) They have only one X-chromosome
(d) Commonly they have less fat

41st B.P.S.C. (Pre) 1996

Ans. (c)

Colour blindness is inherited in an X-linked recessive pattern. The genes are located on the X-chromosome which is one of the two sex chromosomes. In males (who have only one X-chromosome) one genetic change in each cell is sufficient to cause the condition. Males are affected by X-linked recessive disorders much more frequently than females (who have two X-chromosomes) in which a genetic change would have to occur on both copies of the chromosome to cause the disorder. A characteristic of X-linked inheritance is that father cannot pass X-linked traits to their sons.

VI. Deficiency, Non-Contagious

and Other Diseases

1. **Which of the following pairs are correctly matched?**

1. **Anorexia** : **Sleep disorder**
2. **Insomnia** : **Eating disorder**
3. **Dyspnoea** : **Shortness of breath**
4. **Anosmia** : **Partial or full loss of smell**

Select the correct answer using the codes given below.

- (a) Only 1 and 2 (b) Only 2 and 3
(c) Only 3 and 4 (d) All of the above

69th B.P.S.C. (Pre) 2023

Ans. (c)

Anorexia is associated with Eating Disorder. Insomnia is associated with Sleep Disorder. Dyspnoea is associated with Shortness of breath and Anosmia is associated with Partial or full loss of smell. Hence only option 3 and 4 are correct.

2. The malfunctioning of thyroid gland is due to the deficiency of :

- (a) Vitamin A
- (b) Calcium
- (c) Iodine
- (d) Iron
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

The malfunctioning of thyroid gland is due to the deficiency of iodine. Goitre is an abnormal enlargement of thyroid gland, the most common cause of which is lack of iodine in diet. Deficiency of iodine leads to decreased production of thyroxine and triiodothyronine hormones from the thyroid gland which are necessary to regulate metabolic rate and other vital body functions.

3. Itai-itai disease is caused by

- (a) mercury
- (b) iron
- (c) cadmium
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

Itai-itai disease is caused by cadmium exposure. Itai-itai disease is characterised by Osteomalacia with severe bone pain and is associated with renal tubular dysfunction.

4. Methemoglobinaemia is caused by the drinking of water contaminated with

- (a) nitrate
- (b) cadmium
- (c) phosphate
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

Methemoglobinemia (MetHb) is a blood disorder in which an abnormal amount of methemoglobin is produced. The most common cause of methemoglobinemia is the ingestion or inhalation of oxidizing agents such as nitrates or nitrites.

5. Which one of the following is used in the treatment of blood cancer?

- (a) Iodine-131
- (b) Sodium-24
- (c) Phosphorus-32
- (d) Cobalt-60
- (e) None of the above / More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

Phosphorus-32 and cobalt-60 both radioisotopes are used to control blood cancer (leukaemia).

6. The radiations used in the treatment of muscle ache are :

- (a) Infrared
- (b) Microwave
- (c) UV
- (d) X-ray
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Infrared radiations are used in the treatment of muscle ache. Infrared light is one of several innovative therapies that are being trialed for the management of patients with acute or chronic pain. Infrared light radiation helps cells regenerate or repair themselves. It has been shown to be effective in the treatment of muscle or joint pain, including muscle spasm and stiffness.

7. The incubation period of hepatitis B virus is

- (a) 42-56 days
- (b) 30-180 days
- (c) 15-20 days
- (d) one week
- (e) None of the above / More than one of the above

B.P.S.C. CDPO 2018

Ans. (b)

The incubation period of the hepatitis B virus ranges from 30 to 180 days. The virus may be detected within 30 to 60 days after infection and can persist and develop into chronic hepatitis B, especially when transmitted in infancy or childhood.

8. Oncogenes are associated with –

- (a) Tuberculosis
- (b) Hepatitis
- (c) Cancer
- (d) Typhoid

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

An oncogene is a kind of abnormal gene that predisposes cells to develop into cancers. The first oncogene ever identified is termed Sarcoma (SRC) – (Pronounced 'Sarc' as it is short for Sarcoma), discovered in 1970. It is a component of cancer-causing virus in chickens, known as the rous sarcoma virus.

9. Which of the following can cause cancer as well as cure it depending upon its intensity and use?

- (a) Tobacco
- (b) Alcohol
- (c) Ionized radiation
- (d) Ultraviolet rays

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Ionized radiation is the factor which can cause cancer as well as cure, depending upon its intensity and use. Photon radiation and particle radiation are the types of ionized radiation used in cancer treatment effectively.

10. The Nobel Prize for developing treatment of Parkinson's disease was given to:

- (a) Arvid Carlsson
- (b) John F. Enders
- (c) Robert B. Laughlin
- (d) Walter Kohn
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Arvid Carlsson (1923 - 2018) was a Swedish neuropharmacologist, whose discoveries about the brain led to the development of drugs for Parkinson's disease. He is best known for his work with the neurotransmitter dopamine and its effects in Parkinson's disease, for which he was awarded the Nobel Prize in Physiology or Medicine in 2000, together with Eric Kandel and Paul Greengard.

11. Lack of which substance causes diabetes in a person?

- (a) Glycine
- (b) Haemoglobin
- (c) Histamine
- (d) Insulin

44th B.P.S.C. (Pre) 2000

Ans. (d)

Insulin is a hormone made in the pancreas, that allows our body to use sugar (glucose) from carbohydrates in the food that we eat for energy or to store glucose for future use. Insulin helps to keep blood sugar level from getting too high (hyperglycemia) or too low (hypoglycemia). Lack of Insulin in the body causes diabetes in a person. Diabetes patients have excess urination problem called polyuria.